FINAL PROJECT • MOBILE APPLICATION DEVELOPMENT

PantryWise

Smart Anti-Waste Kitchen & AI Recipe Assistant

Transforming Expiring Pantry Items into Meals with Gemini AI & Room Persistence

PRESENTER

Android Development Team

TECH STACK

Android • Jetpack Compose • Kotlin Coroutines • Room DB • Gemini AI

SYSTEM OVERVIEW

Project Description

A modern Android application reducing household food waste through intelligent inventory tracking and AI.

Smart Expiry Tracking

Color-coded badges highlighting urgent items expiring soon.

AI Recipe Generator

Instant Gemini AI meal suggestions generated via Kotlin Coroutines.

Offline Persistence

Room Database storing inventory items and favorite recipes offline.

Multi-Screen Flow

Jetpack Compose BottomBar linking Pantry, Saved, and AI Advisor.

PROBLEM & SOLUTION

Issues and Motivations

Addressing household food waste through modern mobile engineering.

The Issues

Massive Food Waste: Perishable groceries expire unnoticed in household refrigerators.

Kitchen Decision Fatigue: Uncertainty over what to cook with mismatched leftover ingredients.

Financial Loss: Unnecessary food delivery orders resulting in discarded fresh produce.

The Motivations

Real-Time Visibility: Color-coded expiration alerts prioritize what to cook first.

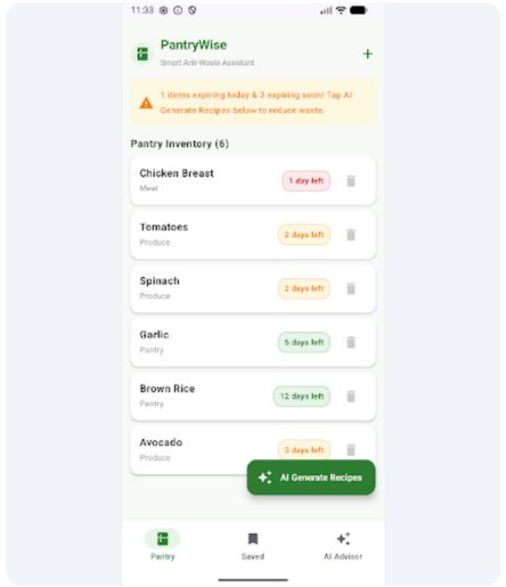
Generative AI Assistance: Gemini AI generates tailored recipes from available items.

Android Engineering: MVVM architecture with Room persistence and Jetpack Compose.

UI / UX ARCHITECTURE

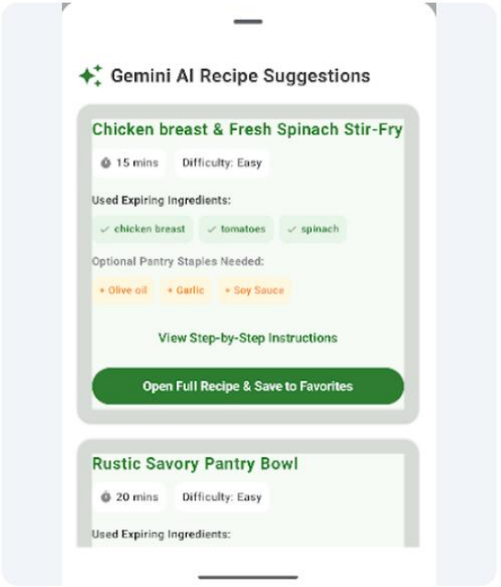
Screen Design & App Interface

Material 3 Android user interface screens implementing anti-waste workflows.



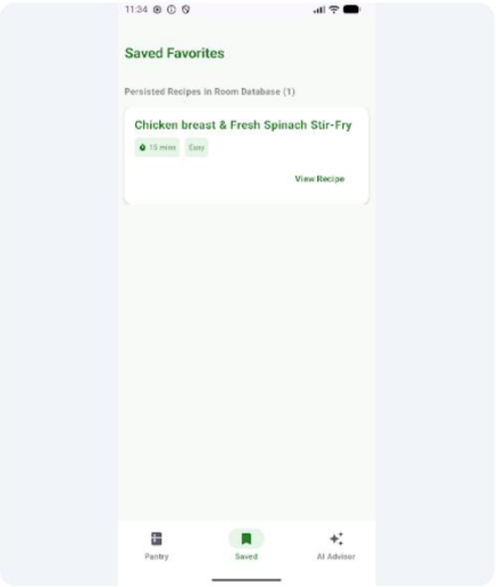
Pantry Inventory

Expiration badges & AI generate FAB.



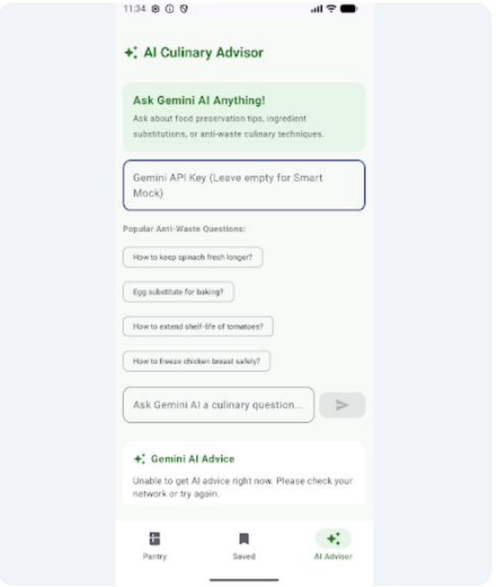
AI Recipes

Generated cards with ingredient chips.



Saved Favorites

Room DB persisted recipes list.



AI Advisor

Interactive Q&A for food preservation.

PRESENTATION SCRIPT

Live Demonstration Flow

Step-by-step feature execution verifying mandatory technical requirements.

STEP 1

Inventory Review

View expiring item badges loaded from Room DB.

STEP 2

Generate Recipes

Trigger Gemini AI prompt via Kotlin Coroutines.

STEP 3

Explore Cards

Inspect prep time, difficulty, and ingredient chips.

STEP 4

Save Favorite

Persist recipe to local Room Database.

STEP 5

Consult AI

Ask anti-waste advice in AI Culinary Advisor.

REQUIREMENT VERIFICATION MATRIX

- **Multiple Screens:** 3 NavDestinations
- **Data Persistence:** Room DB DAO
- **External Connectivity:** Coroutines
- **AI Integration:** Gemini 1.5 REST